

Accurate and high-resolution mapping of shrub coverage is critical for determining fuel load, modelling fire behavior and planning risk reduction studies, especially in fire-sensitive ecosystems as Mediterranean. However, the fact that shrubs exhibit spectral characteristics similar to trees and bare soil, as well as the limited pixel accuracy of training label, limits the success of current remote sensing methods. In this study, an automatic shrub classification workflow was developed that integrates NAIP aerial images with 1 m resolution, Lidar- derived Canopy Height Model (CHM), Terrestrial Laser Scanning (TLS), Segment Anything Model (SAM) and XGBoost algorithm.

The method was applied in six study sites in California, USA with different ecological and structural characteristics. The approximately circular shrub labels produced from TLS-based land observations were converted to pixel-sensitive boundaries using SAM; hard-negative samples belonging to tree and rock/ground classes were automatically extracted via CHM thresholds. Out of 55 candidate characteristics consisting of NAIP spectral bands, vegetation indicates, canopy height derivatives and multi-scale texture metrics, the most distinctive 20 characteristics were selected and the classifier was trained with multi-purpose Optuna optimization.

The final XGBoost model yielded an F1-score of 0.823, precision of 0.753, recall of 0.907, and AUC-ROC of 0.772 when evaluated on spatially independent validation datasets. Feature importance analysis revealed that canopy height texture metrics computed at multiple spatial scales played a dominant role in model predictions, showing that structural attributes offer greater discriminative power than raw spectral bands.

The results also revealed that SAM-enhanced labels provide a more balanced and reliable classification compared to height threshold-based labels only and deep learning approaches that require more data. This developed approach provides a scalable and interpretable solution for high-resolution bush mapping under limited labeled data conditions.

Keywords: Shrub mapping, forest fire fuels, Segment Anything Model, XGBoost, LIDAR, canopy height Model, remote sensing

